

AKEWAK JEBA (AKI)

PROFILE

Results-driven technology professional with 7+ years of experience spanning data science, artificial intelligence, machine learning, software development, research computing, and IT support. Experienced in developing analytical solutions, software tools, data platforms, and scalable systems that support research, innovation, and decision-making.

Strong hands-on experience with Python, R, machine learning, NLP, cloud/HPC computing, Docker, Kubernetes and reproducible software development. Currently developing AI-agent and Retrieval-Augmented Generation (RAG) solutions integrating LLMs, computer vision, external APIs, structured data sources and retrieval tools.

Passionate about leveraging data, technology, and AI-driven solutions to generate insights, improve processes, support strategic decision-making, and create meaningful impact across diverse industries and domains.

CONTACT

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AI & MACHINE LEARNING ENGINEER | DATA SCIENTIST |
AI SOLUTION DEVELOPER

WORK EXPERIENCE

PROJECT RESEARCHER & DATA SCIENTIST

University of Turku | April 2023 – July 2026

- Develop and maintain open-source software packages, scalable data pipelines and reproducible workflows.
- Design machine learning, statistical modeling, and data integration solutions for complex biological and healthcare-related datasets.
- Designed scalable analytical workflows leveraging cloud and HPC infrastructure to support machine learning, data integration, and large-scale scientific data analysis.
- Designed and maintained software systems and data services supporting international scientific collaborations.
- Contribute to large-scale digital humanities initiatives involving harmonization and analysis of more than one million cultural metadata records.
- Develop and maintain Software packages and reproducible research tools used by academic and research communities.
- Developing AI-agent and Retrieval-Augmented Generation (RAG) systems, integrating LLMs with external APIs, structured data sources and retrieval tools.
- Support project planning, stakeholder communication, technical documentation, and coordination of multidisciplinary research activities.

FULL-TIME, INDEPENDENT CONSULTANT

Appen | Jan 2020 – Mar 2023

- Performed image classification, semantic segmentation, and object detection, delivering insights for computer vision applications.
- Conducted query-result analysis to assess relevance, identify key events, and classify entities to enhance information retrieval accuracy.
- Analyzed social media content for sentiment detection, leveraging NLP techniques to classify and categorize sentiments in diverse datasets.
- Conducted document classification across various domains to support automated content organization and retrieval processes.
- Executed ad hoc data collection, information correction, content moderation, and survey research, ensuring data integrity and compliance with quality standards.
- Developed and optimized workflows for data annotation and model training to improve the efficiency of machine learning pipelines.
- Provided actionable recommendations based on data analysis findings to enhance decision-making processes for clients.

RESEARCH ASSISTANT

Tampere University | Sep 2019 – Dec 2019

- Conducted in-depth research and literature reviews on the assigned subject area to identify knowledge gaps and refine research objectives.
- Developed graphical models and generative processes, introducing novel inference methods to enhance model accuracy and interpretability.
- Designed and implemented machine learning algorithms tailored for analyzing social media data, focusing on sentiment analysis, trend detection, or community identification.
- Collaborated with research team members to refine methodologies and validate results.

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SKILLS

Programming: Python, R, SQL, PostgreSQL, Bash, JavaScript, Java, PHP

Data Science & AI: Machine Learning, Deep Learning, hine Learning, Deep Learning, Generative AI, LLMs, AI Agents, Retrieval-Augmented Generation (RAG), Statistical Modeling, Predictive Analytics, NLP, Computer Vision, Data Mining, Optimization, Data Visualization

Cloud & MLOps: AWS (S3, EC2, Lambda, SageMaker), Docker, Kubernetes, Linux, CI/CD, HPC

Software Engineering: Git, Software Testing, Research Software Development, Reproducible Workflows, Shiny

Tools: Microsoft Excel (Pivot Tables, Power Query, Data Analysis), PyCharm, RStudio, VS Code, Jupyter Notebook

LANGUAGES

English – Fluent

Finnish – Intermediate (B2)

Afan Oromo, Amharic – Native

CERTIFICATION

Containers in Supercomputing Environments

CSC – IT Center for Science

CSC Computing Environments

CSC – IT Center for Science

Unsupervised Learning, Recommenders, Reinforcement Learning

Coursera

Advanced Learning Algorithms; Supervised Machine Learning: Regression and Classification

Coursera

AI & MACHINE LEARNING ENGINEER | DATA SCIENTIST |
AI SOLUTION DEVELOPER

AI SOFTWARE DEVELOPER

Routa Markkinointi Oy | Sep 2018 – Mar 2019

- Conducted data mining, data cleaning, and exploratory data analysis on raw datasets to prepare for model development.
- Developed machine learning algorithms for automated website categorization, enhancing content organization and retrieval.
- Trained machine learning-based software to perform specific tasks with high accuracy and efficiency.
- Deployed machine learning models into production environments, ensuring smooth integration and scalability.

RESEARCH ASSISTANT

Åbo Akademi University | Sep 2019 – Dec 2019

- Developed AI and machine learning algorithms to model and predict epidemic trends.
- Engaged in full-stack web development.
- Collaborated with a multidisciplinary team to integrate epidemiological models with social media data analytics for comprehensive public health insights.
- Conducted data cleaning and preprocessing to ensure data quality, accuracy, and relevance for machine learning applications.

WEB DEVELOPER

Freelancer | Sep 2011 – Dec 2019

- Write well-designed, testable, efficient code by using web development best practices.
- Create basic website layouts and user interfaces with HTML, CSS, JavaScript.
- Integrate data from various back-end web services and databases.
- Work with stakeholders to gather & refine specifications requirements.
- Create and maintain software documentation.
- Ensure cross-browser compatibility of new features.
- Be responsible for maintaining, expanding, and scaling of the websites.

LEADERSHIP & PROFESSIONAL SERVICE

DEPUTY SECRETARY

CODATA Finland | January 2024 – Present

- Support national and international initiatives promoting open science, FAIR data principles, and scientific collaboration.
- Contribute to coordination of scientific activities, stakeholder engagement, and organizational operations.

SCIENTIFIC COMPUTING AMBASSADOR

CSC | September 2025 – Present

- Promote advanced computing, artificial intelligence, machine learning, and data-intensive research practices within the Finnish research ecosystem.
- Support researchers in adopting high-performance computing, reproducible research, and modern data science methodologies.

PROJECTS

2026	Agentic AI & RAG for Cultural Heritage
2024–2026	Metadata Harmonization and Enrichment Platform
2024	NCBI Resistome Codeathon
2023–2026	Harmonized Finnish National Bibliography
2023–2025	Microbiome Analytics Tools (miaverse)
2023–2025	European Music Metadata Infrastructure

VOLUNTEER EXPERIENCE

The Salvation Army Turku corps

REFERENCE

Available upon request

AKEWAK JEBA (AKI)

AI & MACHINE LEARNING ENGINEER | DATA SCIENTIST | AI SOLUTION DEVELOPER

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE
Åbo Akademi University | 2017 – 2021
Graduate Degree

BACHELOR OF ENGINEERING IN INFORMATION TECHNOLOGY
Metropolia University of Applied Sciences | 2006 – 2011